

“BlazeBot : Automatic Fire Extinguishing Robot”

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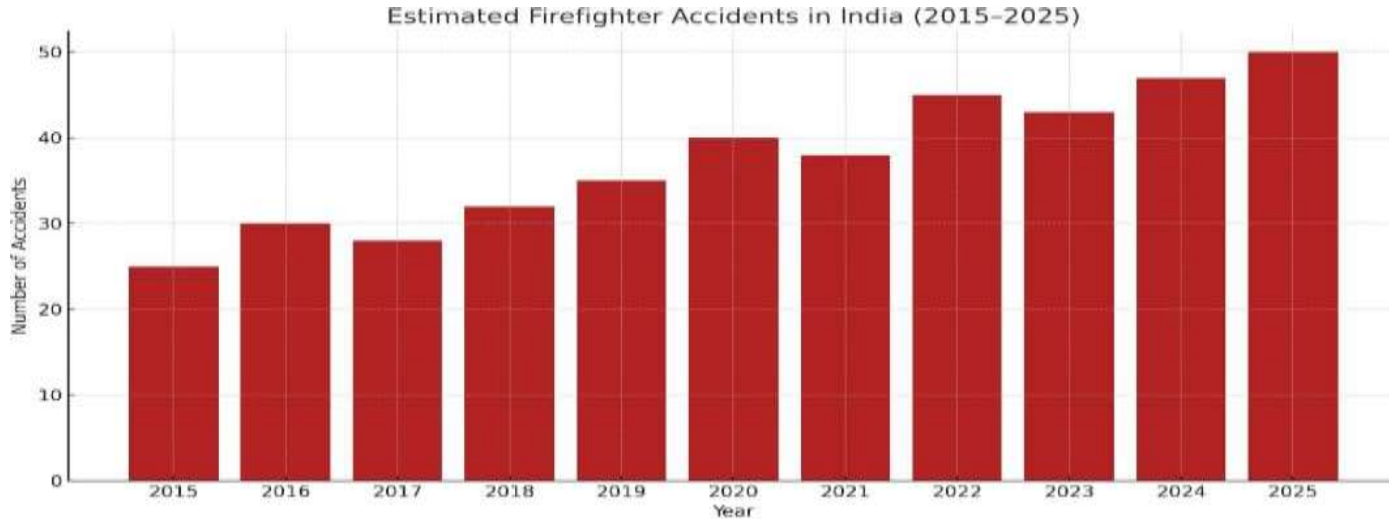


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Motivation:

Fire-fighting is a crucial operation that currently needs to be performed regularly by humans. Fire fighters face risky situations when extinguishing fires and rescuing victims, it is an inevitable part of being a fire fighter. In contrast, a robot can function by itself or be controlled from a distance, which means that fire fighting and rescue activities could be executed without putting fire fighters at risk by using robot technology instead. In other words, robots decrease the need for fire fighters to get into dangerous situations.



Objective:

The main objective of our project is to design and test a robot capable of detecting and locating fire using advanced sensors and to navigate and operate independently for extinguishing building and basement fires, they effectively reduce human risk by replacing firefighters in hazardous environments .



Introduction:

BlazeBot aim to revolutionize fire safety. By utilizing advanced sensors to detect and locate fires, they minimize human risk in hazardous environments. These robots are designed for autonomous navigation and operation in complex terrains, enabling rapid response and efficient fire suppression using various extinguishing mechanisms. By minimizing damage and firefighting costs, these robots enhance cost-effectiveness. Their versatility allows for operation in diverse scenarios, from homes to industrial settings.



Block Diagram:

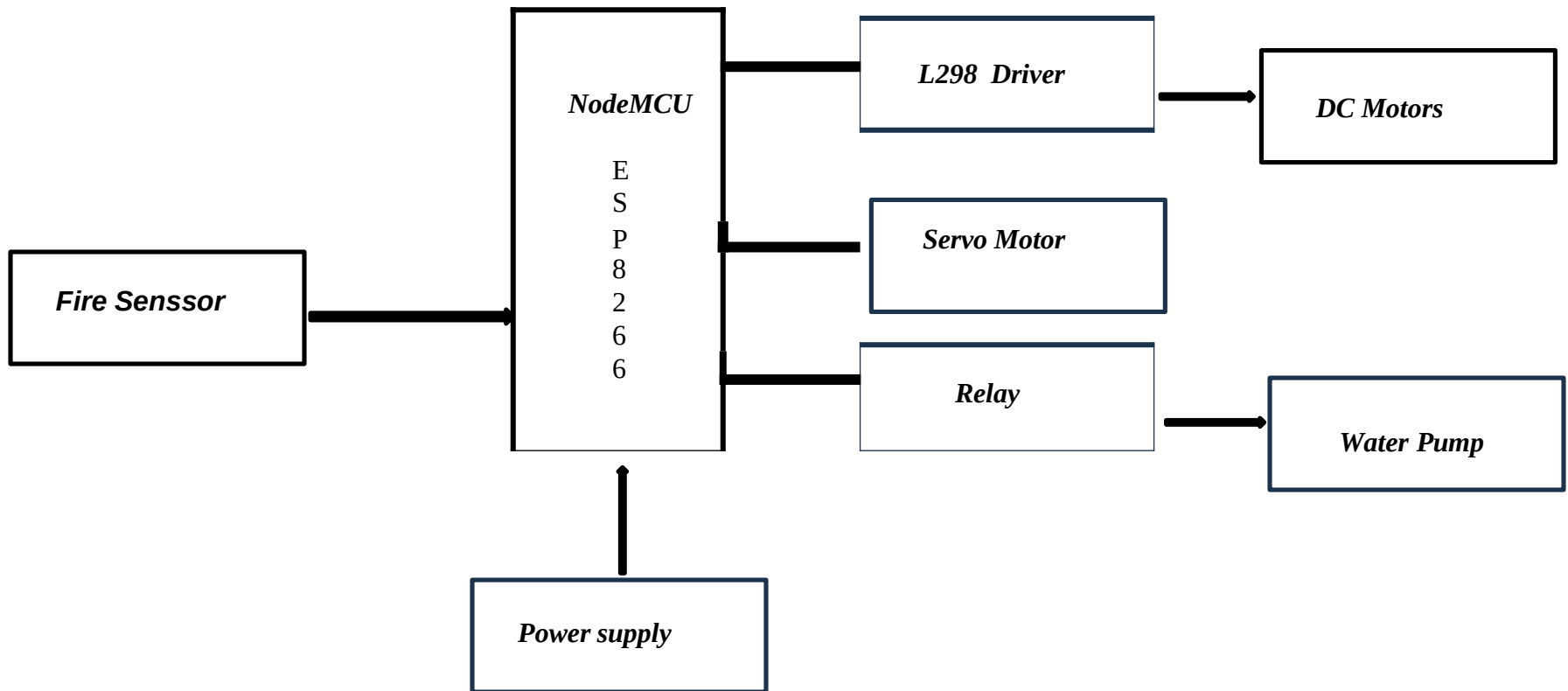


Figure 1 Block Diagram:” BlazeBot : Automatic Fire Extinguishing Robot”

ESP8266 Wifi Module:

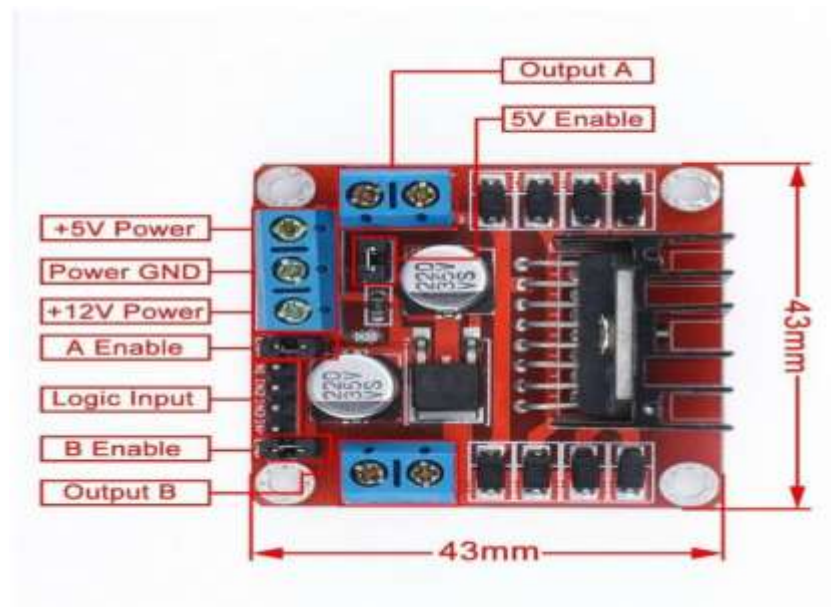
- **Operating Voltage** : works at 3.3V (not tolerant to 5V)
- **GPIO** : Provides upto 17 GPIO pins
- **Memory** : 64KB RAM (for instructions)
96KB (for data)
512KB to 4MB (Program storage)
- **Communication** : UART (Serial communication)
SPI (High speed data transfer)
I2C (for connecting sensors)
PWM (controls motor speed and brightness of LED's)
ADC (analog sensor input)
- **Power Consumption** : 70 – 200 mA
- **Operating modes** : Station mode (STA) connects to existing n/w
Access point mode (AP) creates its own wifi n/w
Dual mode (STA + AP) functions as both at a time



L298 Motor Server:

The L298 motor driver is a dual H-bridge motor driver IC commonly used to control DC motors .It allows you to control the speed and direction of two motors independently

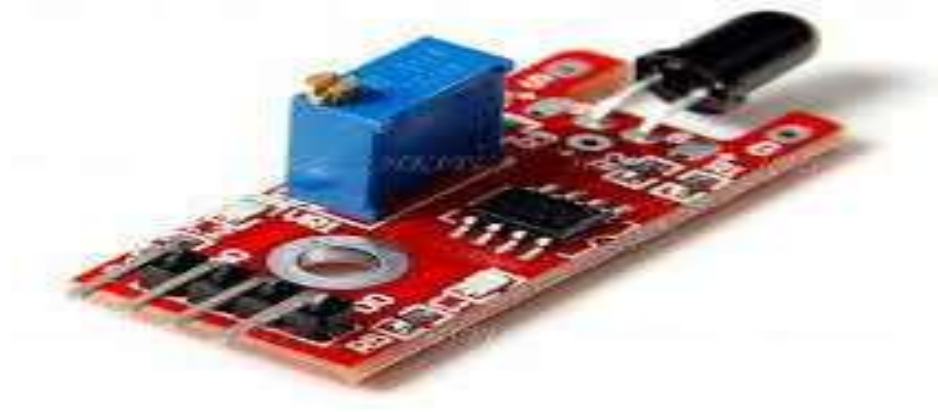
- **Inputs (IN1, IN2, IN3, IN4):** These pins control the direction of the motor.
- **Enable Pins (ENA, ENB):** These pins control whether the motors are enabled and are also used to adjust motor speed via PWM.
- **Motor Terminals (Out1, Out2, Out3, Out4):** These are connected to the motors.
- **Vss:** Power for the logic (often 5V from a microcontroller or external supply).
- **Vs:** Power for the motors (the voltage required by the motors, which can be higher than Vss).
- **GND:** Ground pins for both logic and motor circuits



IR Flame Sensor:

An IR Flame Sensor is a device used to detect the presence of fire or flame by identifying infrared (IR) light emitted by fire. It is commonly used in fire detection systems, safety equipment, and robotics projects.

- **Detection Angle:** Wide detection range, usually 60° to 120°.
- **Range:** Can detect flames from a distance of up to 100 cm or more, depending on the sensor.
- **Digital Output (D0):** High (1) when flame is detected, Low (0) otherwise.
- **Analog Output (A0):** Proportional to the intensity of the detected flame.
- **Adjustable Sensitivity:** Some models have a potentiometer to adjust the sensitivity.
- **Power Supply:** Typically operates at 3.3V to 5V.



BO Motor and Buzzer:

BO Motors (Battery Operated Motors) x 4 with Wheels refers to a set of four DC gear motors, commonly used in robotics and DIY projects to provide movement to wheeled platforms.

- Operating Voltage: Typically 3V to 12V
- Speed Range: 60-300 RPM (varies based on gear ratio and voltage)

A buzzer is an electronic device that produces a sound when electrical energy is applied to it.

- It is commonly used in electronic circuits to indicate
- The status of the system or to alert users of a specific event.
- In the context of our prototype, the buzzer is used to produce an audible alarm when the fire is detected.
- The sound level and frequency of the buzzer can be adjusted by changing the resistance of the circuit.



Resistor and BD140 Transistor:

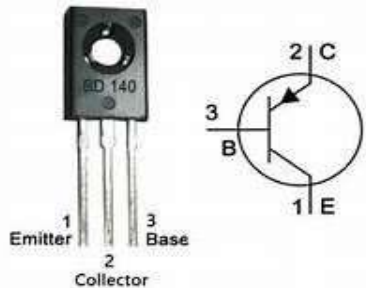
A resistor is a passive electrical component that is used to limit the current flow, provide voltage drops, and divide voltages.

- In this project the resistor protects the transistor by limiting the current flowing to the base and ensures proper switching operation of the BD140 transistor.
- The BD140 is a PNP transistor that acts as a current-controlled device, meaning a small base current controls a larger collector current.
- Here it acts as a switch to control the water pump and buzzer.

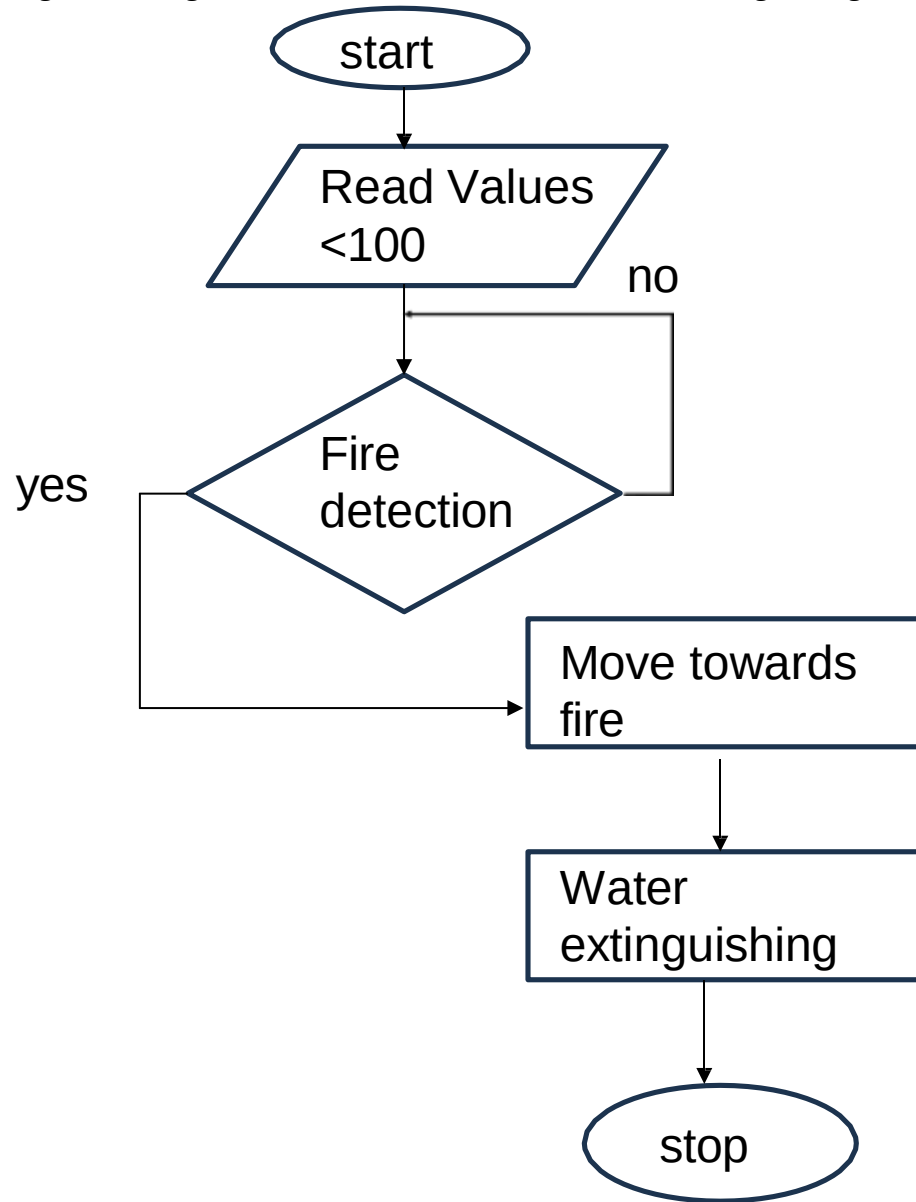
emitter (E) is connected to 3.7V supply.

collector (C) is connected to the positive terminal of the water pump and buzzer

base (B) is connected to the DO pin of the Flame Sensor via a $1K\Omega$ resistor



Design Flow: Figure 2 Design Flow : "BlazeBot : Automatic Fire Extinguishing Robot"



Result:



Figure 3 When fire is detected. Buzzer is activated and water is extinguished.



Figure 4 Robot movement is controlled through mobile app

Advantages:

- The fire detecting robot helps in following ways:
- To detect the exact direction of the fire source.
- Capability of sensing accurately with increased flexibility.
- Reduce human effort.
- Reliable and economical.
- Not sensitive to weather conditions.

Applications:

- Can be used in server rooms.
- Extinguishes fire where probability of explosion is high.
- Usable in Power plant control rooms, Captain bridges, Flight control centres.
- Disaster area monitoring and rescue.

Future Scope:

Future scope for Fire-extinguishing robot is quite promising, with several exciting avenues for development.

Enhanced autonomy: Swarm robotics, AI-powered decision making, and improved navigation.

- Advanced capabilities: Better sensors, versatile mobility, and multi-functional tools.
- Integration with humans: Remote control, teleoperation, and human-robot collaboration.
- Specialized applications: Wildfire fighting, industrial fire suppression, and search and rescue.
- Predictive maintenance and prevention: Early fire detection and fire safety inspections.
- Addressing challenges: Reliability, ethical considerations, and standardization.

Conclusion:

This project successfully demonstrated the feasibility of developing a basic fire-extinguishing robot using NodeMCU. The robot effectively navigated the environment, detected simulated fire sources, and executed extinguishing actions. While limitations such as limited range and basic obstacle avoidance exist, this project serves as a valuable foundation for further development. Future work will focus on enhancing autonomy, improving fire detection capabilities, and expanding the robot's functionalities to address real-world firefighting challenges. This project highlights the potential of NodeMCU based boards for developing innovative and cost-effective solutions for fire safety and emergency response.

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Thank you!